

Preliminaries—nonincreasing virtual utility

- General IPVM model, not necessarily symmetric and without assuming the regularity condition.
- $X_i = [\underline{x}_i, \bar{x}_i]$ type space agent i , $x_i \in X_i$.
- $X = \prod_{i=1}^I X_i$ space of type profiles, $x \in X$.
- $F_i(x_i)$ and $f_i(x_i)$ are agent i 's cdf and respective pdf;
 $(\forall x_i \in X_i) f_i(x_i) > 0$.
- $f(x) = f_1(x_1) \times f_2(x_2) \times \cdots \times f_I(x_i)$.

Issue

- Seller's expected revenue (SER): $\sum_{i=1}^I \int_{\underline{x}_i}^{\bar{x}_i} v_i(x_i) Q_i(x_i) f_i(x_i) dx_i$.
- The objective function, SER, is maximized by allocating the object to the bidder with the highest, non-negative, virtual valuation. Under regularity, this yields allocation rules $Q_i(x_i)$ that are monotone and thus satisfy IC.
- Issue: when $v_i(x_i)$ violates regularity—that is to say, v_i is not weakly increasing—the $Q_i(x_i)$ obtained as described above may violate IC.
The solution obtained in the regular case does not satisfy IC in the absence of regularity.
- To address this issue, Myerson generalizes the “ironing” procedure in Mussa and Rosen (1978). For an interesting application see Loertscher and Muir (2025).

Approach

- Transform the *irregular* virtual utility—that is to say the distribution that generates it—so that
 - regularity is restored and
 - monotone allocations Q_i generate no less revenue against the new distribution than against the original one.
- Roughly, the steps described identify an increasing proxy $\bar{v}_i$ to use as v_i 's surrogate to solve the optimization.
- Technically, the transformation identifies the “multipliers” in the complementary slackness conditions of the implicit linear program.
- Variable of optimization: $\{p_i\}_{i=1}^I, Q_i(x_i) = E_{-i}(p_i(x_i, x_{-i}))$.

Constructing $\bar{v}_i$

- Define $w_i : [0, 1] \rightarrow \mathbb{R}$, by $w_i = v_i \circ F_i^{-1}$.
- Let $W_i(q) = \int_0^q w_i(r) \, dr$. Note that $q, r \in [0, 1]$.
- Let $\bar{W}_i$ be the highest convex function on $[0, 1]$ such that $\bar{W}_i \leq W_i$.
- Define $\bar{w}_i(p) = \frac{d\bar{W}_i(p)}{dp}$ where the derivative is extended right-continuously where necessary.
- Finally, define $\bar{v}_i(x_i) = \bar{w}_i(F_i(x_i))$.

Theorem 1 IPV. In the optimal mechanism the object is sold to the bidder with highest $\bar{v}_i(x_i)$ provided it is nonnegative.

Proof. Define $v_i : [\underline{x}_i, \bar{x}_i] \rightarrow \mathbb{R}$ and $w_i : [0, 1] \rightarrow \mathbb{R}$ by

$$v_i(x_i) = x_i - \frac{1 - F_i(x_i)}{f_i(x_i)}, \quad (\text{virtual utility})$$

$$w_i(q) = F_i^{-1}(q) - \frac{1 - q}{f_i(F_i^{-1}(q))}$$

Note that $F_i : [\underline{x}_i, \bar{x}_i] \rightarrow [0, 1]$ and $w_i = v_i \circ F_i^{-1}$ and define

$$(\forall x_i \in [\underline{x}_i, \bar{x}_i]) \quad V_i(x_i) = \int_{\underline{x}_i}^{x_i} v_i(y) f_i(y) \, dy$$
$$(\forall q \in [0, 1]) \quad W_i(q) = \int_0^q w_i(r) \, dr.$$

- Observation on v_i and w_i

$$(\forall x_i \in [\underline{x}_i, \bar{x}_i]), V_i(x_i) = W_i(F_i(x_i)),$$

$$\begin{aligned} W_i(F_i(x_i)) &= \int_0^{F_i(x_i)} w_i(r) \, dr = \int_{F_i(\underline{x}_i)}^{F_i(x_i)} \left[F_i^{-1}(r) - \frac{1-r}{f_i(F_i^{-1}(r))} \right] f_i(r) \, dr \\ &= \int_{F_i^{-1}(F_i(\underline{x}_i))}^{F_i^{-1}(F_i(x_i))} \left[y - \frac{1-f_i(y)}{f_i(y)} \right] f_i(y) \, dy \\ &= \int_{\underline{x}_i}^{x_i} v_i(y) f_i(y) \, dy = V_i(x_i) \end{aligned}$$

where $F_i(y) = r$; thus $F_i^{-1}(r) = y$ and $dr = f_i(y) \, dy$ are used in the second line.

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- Let $\bar{W}_i = \text{co } W_i$, that is

$$\bar{W}_i(q) = \inf\{\alpha : (q, \alpha) \in \text{co}(\text{epi } W_i)\}.$$

by construction $\bar{W}_i$ is convex and $\bar{W}_i \leq W_i$

See Rockafellar (1970), page 33, Theorem 5.3, and page 36 for definition of convex hull of a function.

Define $\bar{w}_i(q) = \frac{d\bar{W}_i(q)}{dq}$ where the derivative is extended right-continuously if necessary.

- $\bar{w}_i(q)$ is nondecreasing.

- Surrogate virtual utility. Define

$$(\forall x_i \in [\underline{x}_i, \bar{x}_i]) \bar{v}_i(x_i) = \bar{w}_i(F_i(x_i)).$$

$\bar{v}_i(x_i)$ is weakly increasing because $\bar{w}_i$ is weakly increasing and F_i is strictly increasing.

Let $Y(x) = \{i : \bar{v}_i(x_i) = \max_j \{\bar{v}_j(x_j)\} \wedge \bar{v}_i(x_i) > 0\}$ and

$$\bar{p}_i(x_i, x_{-i}) = \begin{cases} \frac{1}{|Y(x)|} & \text{if } i \in Y(x) \\ 0 & \text{otherwise} \end{cases}$$

Note: If $\bar{v}_i(x_i)$ is constant in a neighborhood of x_i , then $\bar{p}_i(x_i, x_{-i})$ does not change with x_i in that neighborhood.

$$\begin{aligned}
E \sum_{i=1}^I T_i &= \int_X \sum_{i=1}^I \overbrace{v_i(x_i)}^{w_i(F_i(x_i))} p_i(x_i, x_{-i}) f_i(x_i) f_{-i}(x_{-i}) \, dx_i \, dx_{-i} \\
&= \int_X \sum_{i=1}^I \bar{v}_i(x_i) p_i(x) f(x) \, dx \quad (\text{term 1}) \\
&\quad + \int_X \sum_{i=1}^I [v_i(x_i) - \bar{v}_i(x_i)] p_i(x) f(x) \, dx \quad (\text{term 2})
\end{aligned}$$

Then to show that $\bar{p} = \{\bar{p}_i\}_{i=1}^I$ maximizes the OF, one must establish two conditions,

1. $(\forall p)$ that satisfies IC, term 2 must be negative, and
2. $\bar{p}$ maximizes term 1 and makes term 2 = 0.

Condition 1

$$\begin{aligned}
 & \int_{\underline{x}_i}^{\bar{x}_i} \int_{X_{-i}} [v_i(x_i) - \bar{v}_i(x_i)] p_i(x_i, x_{-i}) f_i(x_i) f_{-i}(x_{-i}) dx \\
 &= \int_{\underline{x}_i}^{\bar{x}_i} \int_{X_{-i}} [w_i(F_i(x_i)) - \bar{w}_i(F_i(x_i))] p_i(x_i, x_{-i}) f_i(x_i) f_{-i}(x_{-i}) dx \\
 &= \int_{\underline{x}_i}^{\bar{x}_i} \overbrace{Q_i(x_i)}^{\mathcal{U}} \overbrace{[w_i(F_i(x_i)) - \bar{w}_i(F_i(x_i))] f_i(x_i) dx_i}^{\mathcal{V}'} \text{ (integrate)} \\
 &= \left[W_i(F_i(x_i)) - \bar{W}_i(F_i(x_i)) \right] Q_i(x_i) \Big|_{\underline{x}_i}^{\bar{x}_i} \text{ (evaluates to zero)} \\
 &\quad - \int_{\underline{x}_i}^{\bar{x}_i} \left[W_i(F_i(x_i)) - \bar{W}_i(F_i(x_i)) \right] dQ_i(x_i)
 \end{aligned}$$

Using $W_i(1) - \bar{W}_i(1) = 0 = W_i(0) - \bar{W}_i(0)$,

$$\begin{aligned}
& \int_{\underline{x}_i}^{\bar{x}_i} \int_{X_{-i}} [v_i(x_i) - \bar{v}_i(x_i)] p_i(x_i, x_{-i}) f_i(x_i) f_{-i}(x_{-i}) dx \\
&= - \int_{\underline{x}_i}^{\bar{x}_i} [W_i(F_i(x_i)) - \bar{W}_i(F_i(x_i))] dQ_i(x_i) \\
&\leq 0
\end{aligned}$$

By construction $W_i \geq \bar{W}_i$. Hence for any $Q_i(x_i|p_i)$ increasing,

$$\sum_{i=1}^I \int_{\underline{x}_i}^{\bar{x}_i} [W_i(F_i(x_i)) - \bar{W}_i(F_i(x_i))] dQ_i(x_i|p_i) \geq 0$$

Condition 1 is established.

Condition 2

Recall that seller's expected revenue can be written

$$\begin{aligned} E \sum_{i=1}^I T_i &= \int_X \sum_{i=1}^I \bar{v}_i(x_i) p_i(x) f(x) \, dx \\ &\quad - \sum_{i=1}^I \int_{\underline{x}_i}^{\bar{x}_i} \left[W_i(F_i(x_i)) - \bar{W}_i(F_i(x_i)) \right] \, dQ_i(x_i | p_i) \end{aligned}$$

$$\bar{p} \in \arg \max_{\{p_i\}_{i=1}^I} \int_X \sum_{i=1}^I \bar{v}_i(x_i) p_i(x) f(x) \, dx$$

This is so by definition of $\bar{p}$, $\bar{p}_i(x_i) \in Y(x)$, that is to say $\bar{p}$ places full weight on bidders whose virtual utility is

$$\bar{v}_i(x_i) = \max\{0, (\forall 1 \leq j \leq I) \bar{v}_j(x_j)\}.$$

- Since $\bar{W}_i$ is the convex hull of W_i ($\text{co } W_i$), then

$$\bar{W}_i(p) < W_i(p) \implies \frac{d\bar{w}_i(p)}{dp} = 0.$$

Therefore $\bar{v}_i(x_i) = \bar{w}_i(F(x_i))$ and $\bar{p}_i(x_i|\bar{p})$ must be constant in a neighborhood of $x_i = F_i^{-1}(p)$. This implies that

$$\int_{\underline{x}_i}^{\bar{x}_i} [W(F_i(x_i)) - \bar{W}_i(F_i(x_i))] d\bar{Q}_i(x_i|\bar{p}) = 0$$

This establishes condition 2. QED

graphs

Observation, $v_i(x_i)$ is not increasing

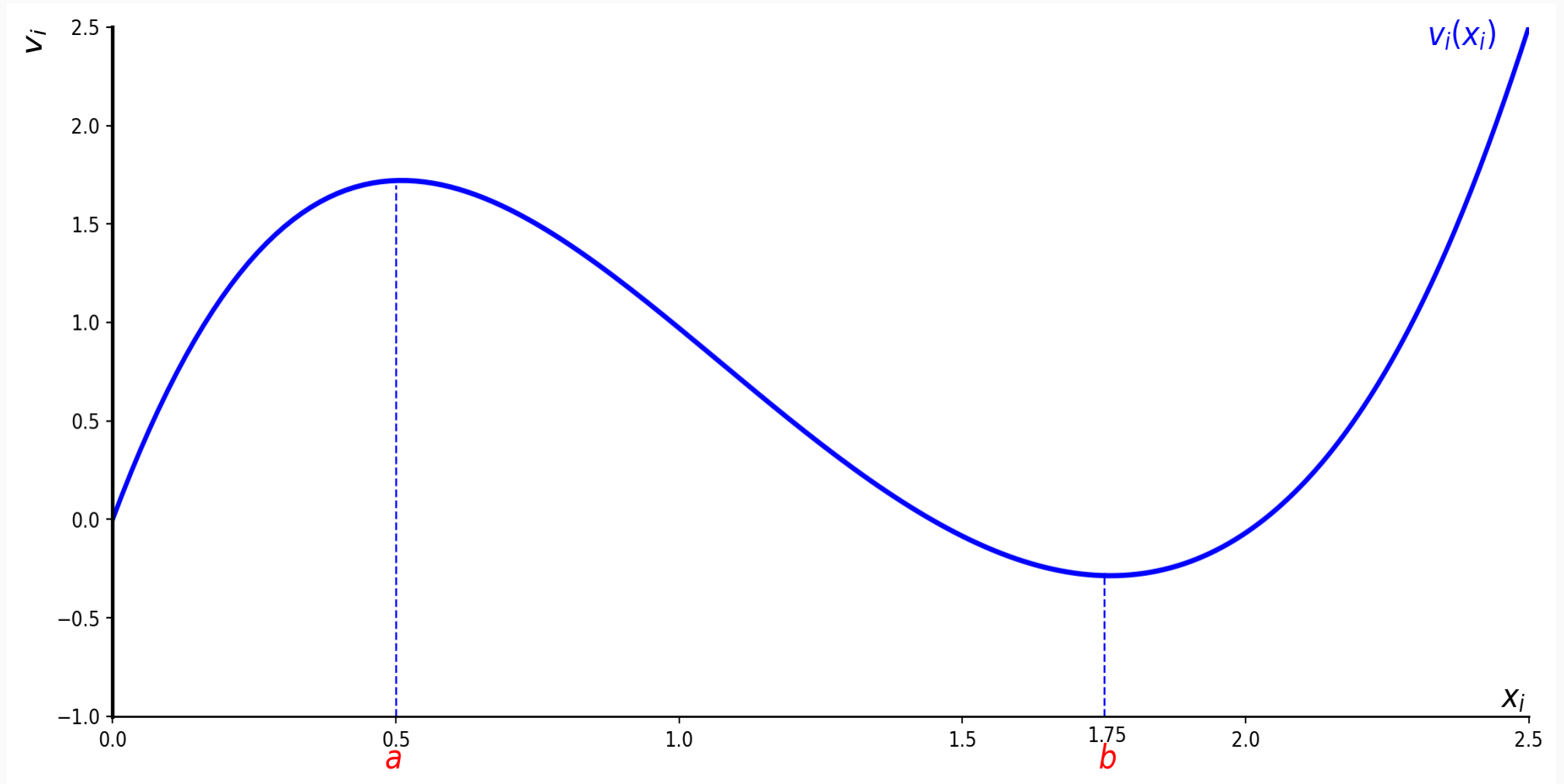


Figure 1

- Given the non-regular $v_i(x_i)$ in Figure 1, let Q'_i be a BIC, DRM. Define

$$Q_i(x_i) = \begin{cases} Q'_i(x_i) & \text{if } x_i \in [0, a) \cup (b, 2.5] \\ \frac{\int_a^b Q'_i(z) f_i(z) dz}{F_i(b) - F_i(a)} & \text{if } x_i \in [a, b] \end{cases}$$

Thus, $(\forall x_i \in [a, b]), Q_i(x_i)$ is a constant

$$Q_i(x_i) = \frac{\int_a^b Q'_i(z) f_i(z) dz}{F_i(b) - F_i(a)} = E_{x_i} [Q'_i(x_i) \mid x_i \in [a, b]].$$

Q_i has two useful properties, (a) it generates weakly more revenue than Q'_i provided $v_i(x_i)$ is non-increasing in $[a, b]$, and (b) it is a BIC direct revelation mechanism. See Fu (2017).

(a) revenue claim

$$\begin{aligned} & \int_a^b Q_i(x_i) v_i(x_i) f_i(x_i) \, dx_i \\ &= E_{x_i} [Q'_i(x_i) \mid x_i \in [a, b]] \int_a^b v_i(x_i) f_i(x_i) \, dx_i \\ &= E_{x_i} [Q'_i(x_i) \mid x_i \in [a, b]] E_{x_i} [v_i(x_i) \mid x_i \in [a, b]] (F_i(b) - F_i(a)) \\ &\geq E_{x_i} [Q'_i(x_i) v_i(x_i) \mid x_i \in [a, b]] (F_i(b) - F_i(a)) \\ &= \int_a^b Q'_i(x_i) v_i(x_i) f_i(x_i) \, dx_i \end{aligned}$$

- Harris inequality.
- Q_i is BIC DRM

(b) Q_i is a BIC DRM

- $Q_i(x_i)$ is non-decreasing because Q'_i is non-decreasing.
- To show it is a mechanism, one must construct for each i , $p_i(x_i, x_{-i})$ so that $Q_i(x_i) = E_{x_{-i}} p_i(x_i, x_{-i})$ and $(\forall x \in X) \sum_{i=1}^I p_i(x) \leq 1$.
Alternatively, let $\{p'_i\}_{i=1}^I$ be the allocation generating $\{Q'_i\}_{i=1}^I$.
 - if i reports $x_i \notin [a, b]$, the $p(x_i, x_{-i}) = p'_i(x_i, x_{-i})$.
 - if i reports $x_i \in [a, b]$, randomly select $x'_i \in [a, b]$ using $\frac{f_i(z)}{F_i(b) - F_i(a)}$ and allocate $p(x_i, x_{-i}) = p'(x'_i, x_{-i})$.
 - Or use Border (1991).
- Jointly (a) and (b) imply that attention can be restricted to mechanisms with flat $Q_i(x_i)$ on $x_i \in [a, b]$.

- Since $Q_i(x_i)$ is constant on $[a, b]$, its revenue against $v_i(x_i)$ equals its revenue against $\bar{v}_i([a, b]) := E_{x_i} [v_i(x_i) | x_i \in [a, b]]$.

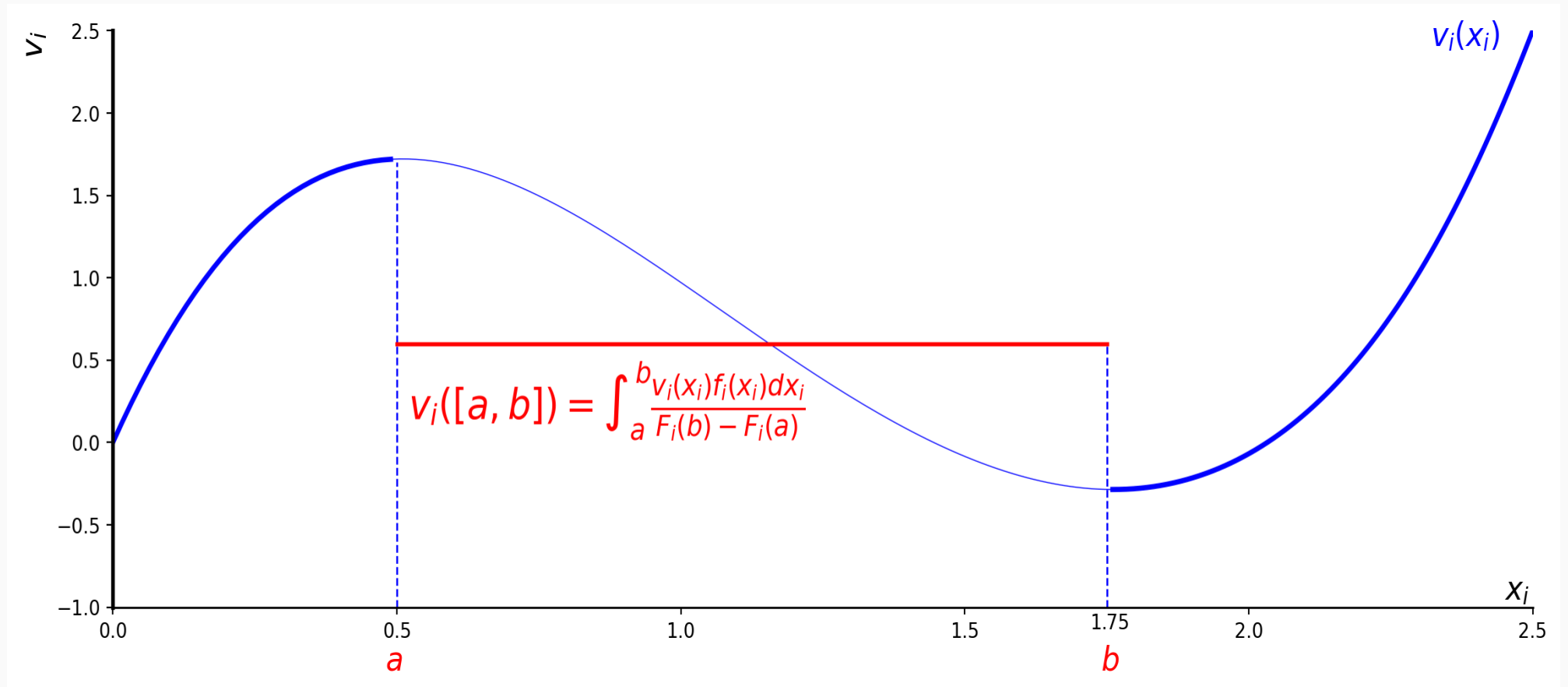


Figure 2

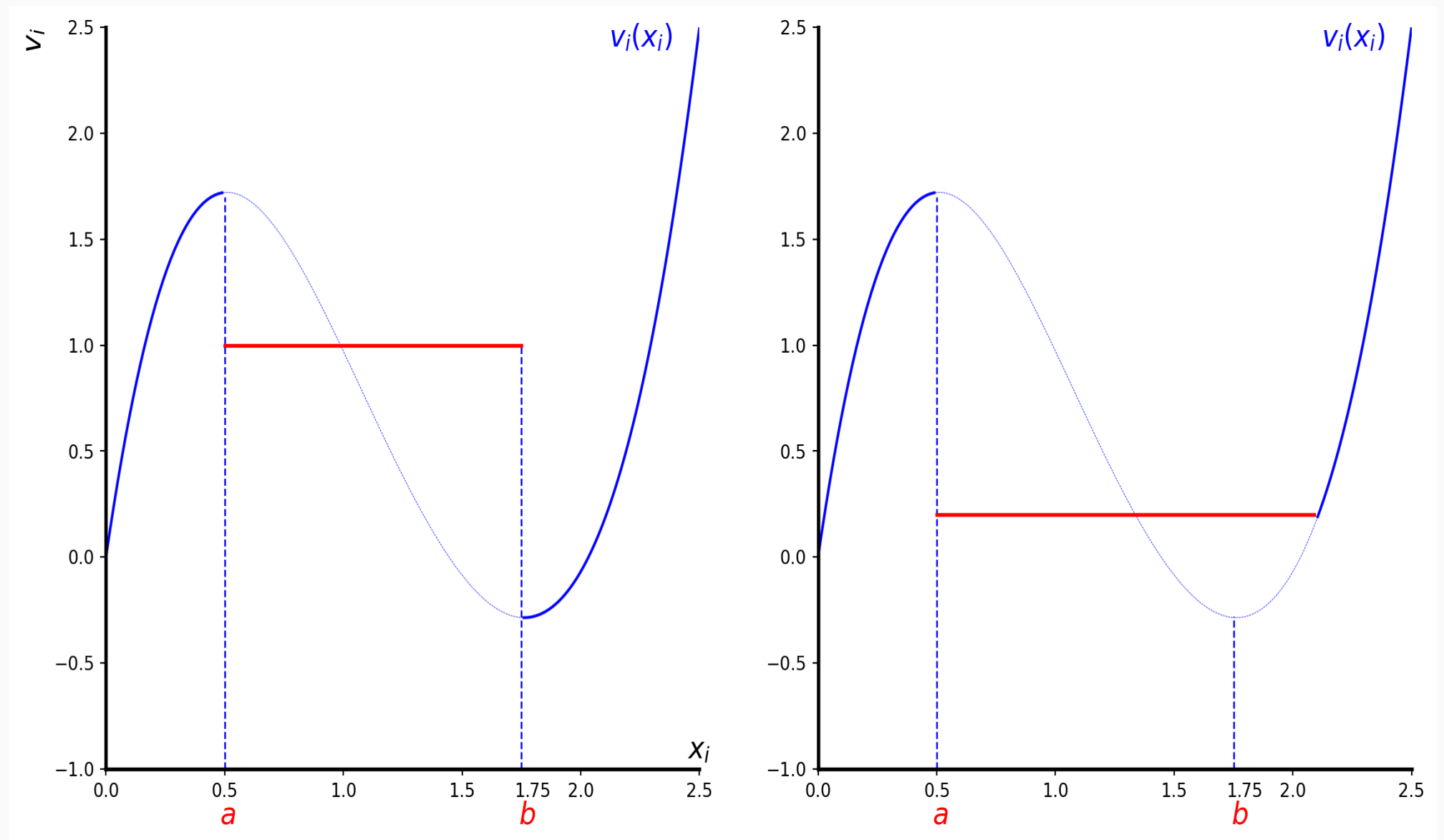


Figure 3

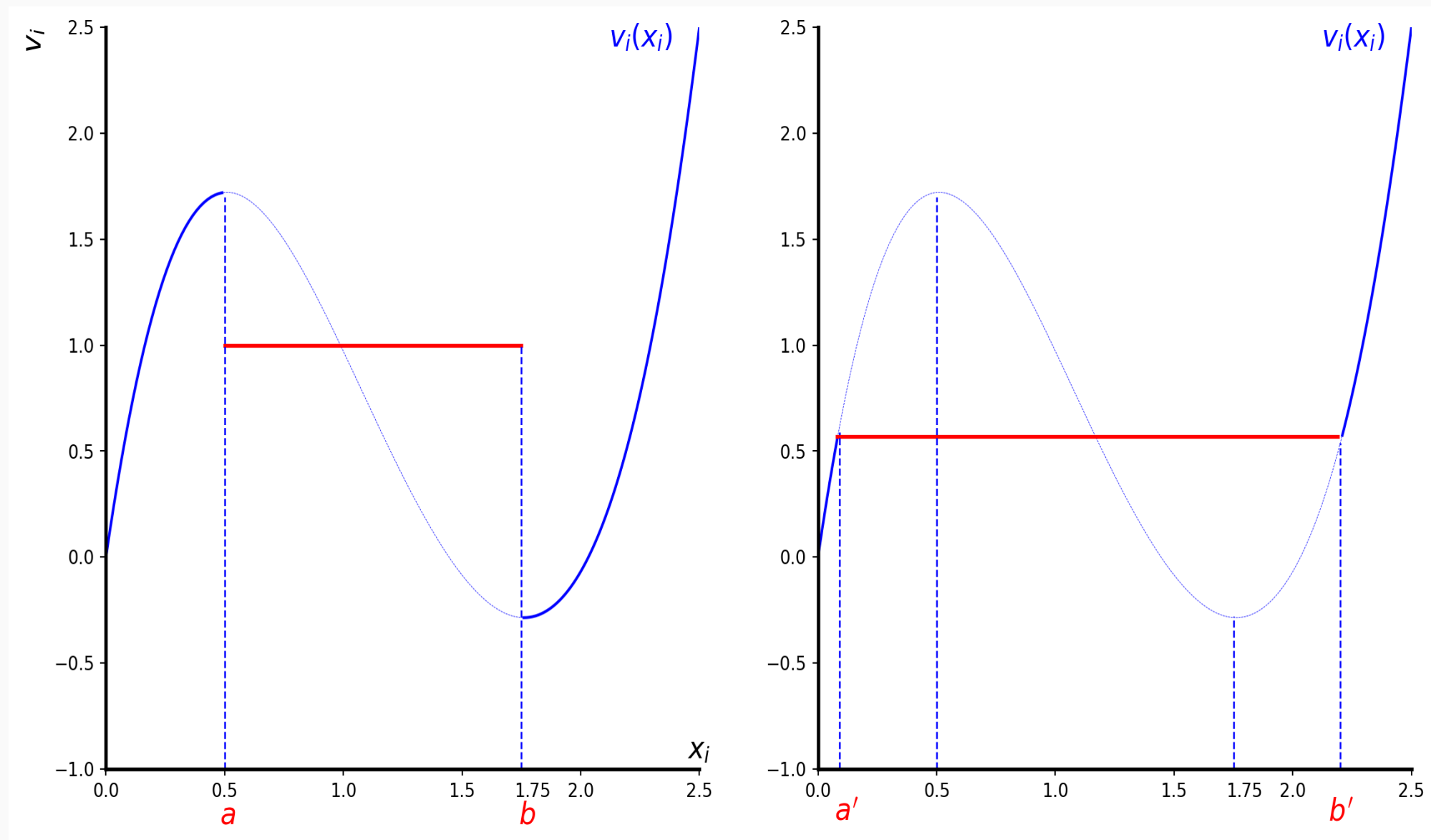


Figure 4

The extent of ironing

- How is $[a', b']$ determine? Is $a' = \underline{x}_i, b' = \bar{x}_i$ valid?

For any $[a, b] \subseteq X_i$, let

$$\bar{v}_i([a, b]) := E_{x_i} [v_i(x_i) | x_i \in [a, b]] = \int_a^b \frac{v_i(x_i) f_i(x_i) dx_i}{F_i(b) - F_i(a)}.$$

Ironing is extended as much as possible provided v_i remains decreasing in the ironed region. This leads to two conditions,

$$v_i(a') \leq \bar{v}_i([a', b']) \leq v_i(b'). \quad (1)$$

$$\forall [c, d] \subseteq [a', b'], \quad v_i(c) \geq \bar{v}_i([c, d]) \geq v_i(d). \quad (2)$$

Define $w_i = v_i \circ F_i^{-1}$. Then

$$w_i(q) = v_i(F_i^{-1}(q)).$$

For $q = F_i(x_i)$,

$$W_i(q) = \int_0^{F_i(x_i)} w_i(r) \, dr = \int_{\underline{x}_i}^{x_i} v_i(y) f_i(y) \, dy$$

$$\left. \frac{dW_i(q)}{dq} \right|_{F_i(x_i)=q} = w_i(F_i(x_i)) = v_i(x_i).$$

Restating the extent of ironing

Let $W_i = \frac{dW_i}{dq}$.

Inequality 1

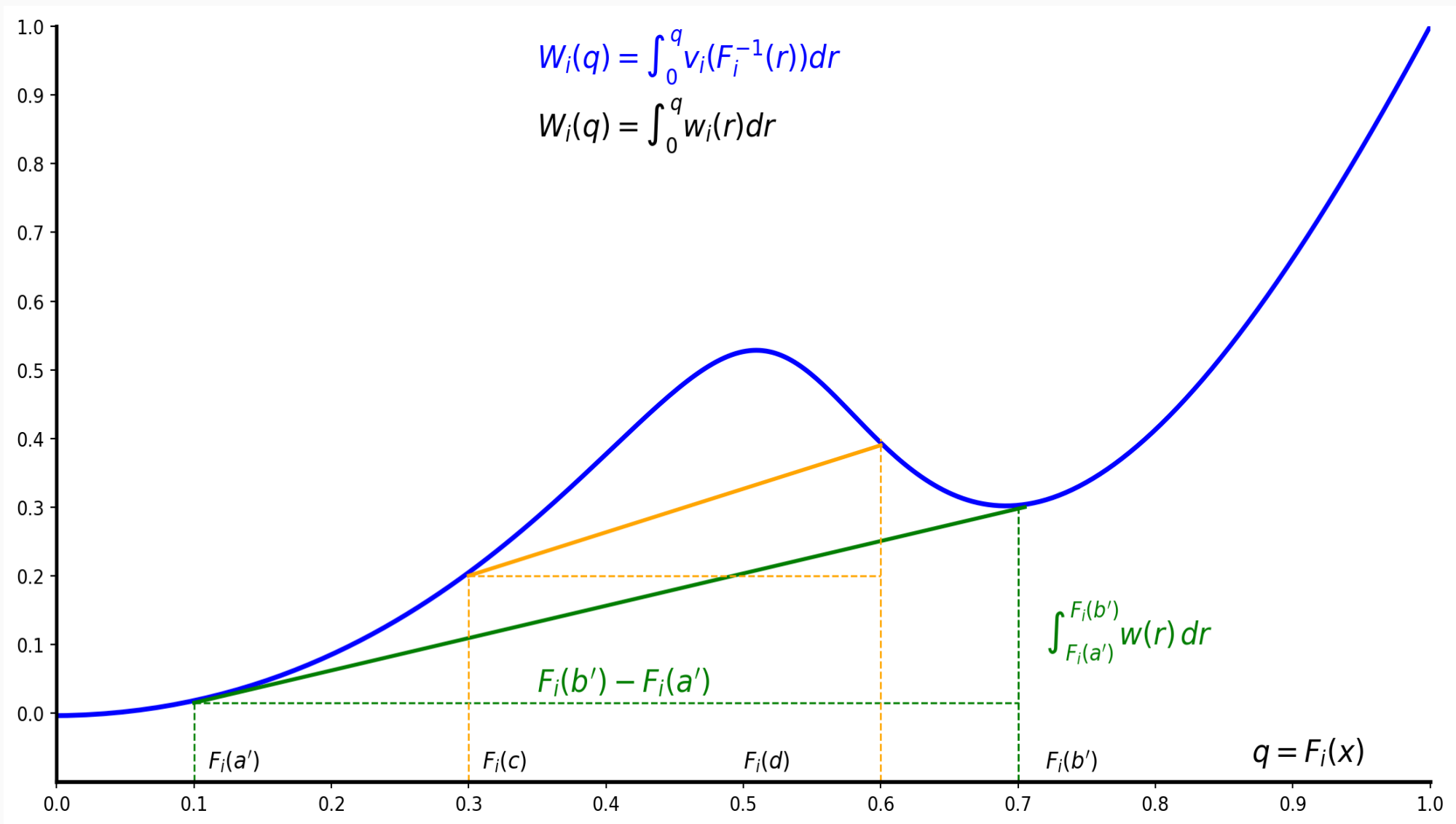
$$\begin{aligned} v_i(a') &\leq \bar{v}_i([a', b']) \leq v_i(b') \\ W_i' \Big|_{F_i(a')=q} &\leq \bar{v}_i([a', b']) \leq W_i' \Big|_{F_i(b')=q} \end{aligned}$$

Inequality 2

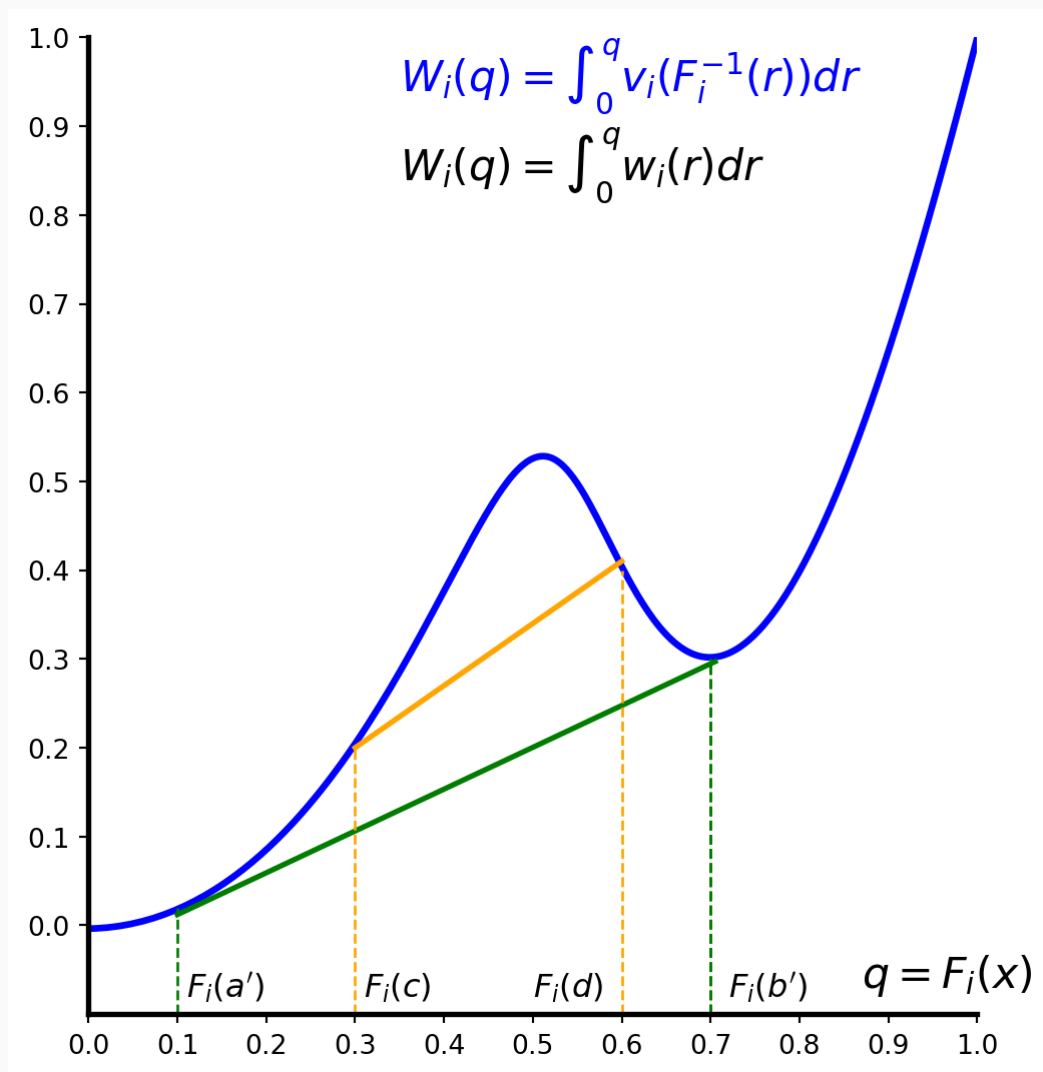
$\forall [c, d] \subseteq [a', b'],$

$$\begin{aligned} v_i(c) &\geq \bar{v}_i([c, d]) \geq v_i(d) \\ W_i' \Big|_{F_i(c)=q} &\geq \bar{v}_i([c, d]) \geq W_i' \Big|_{F_i(d)=q} \end{aligned}$$

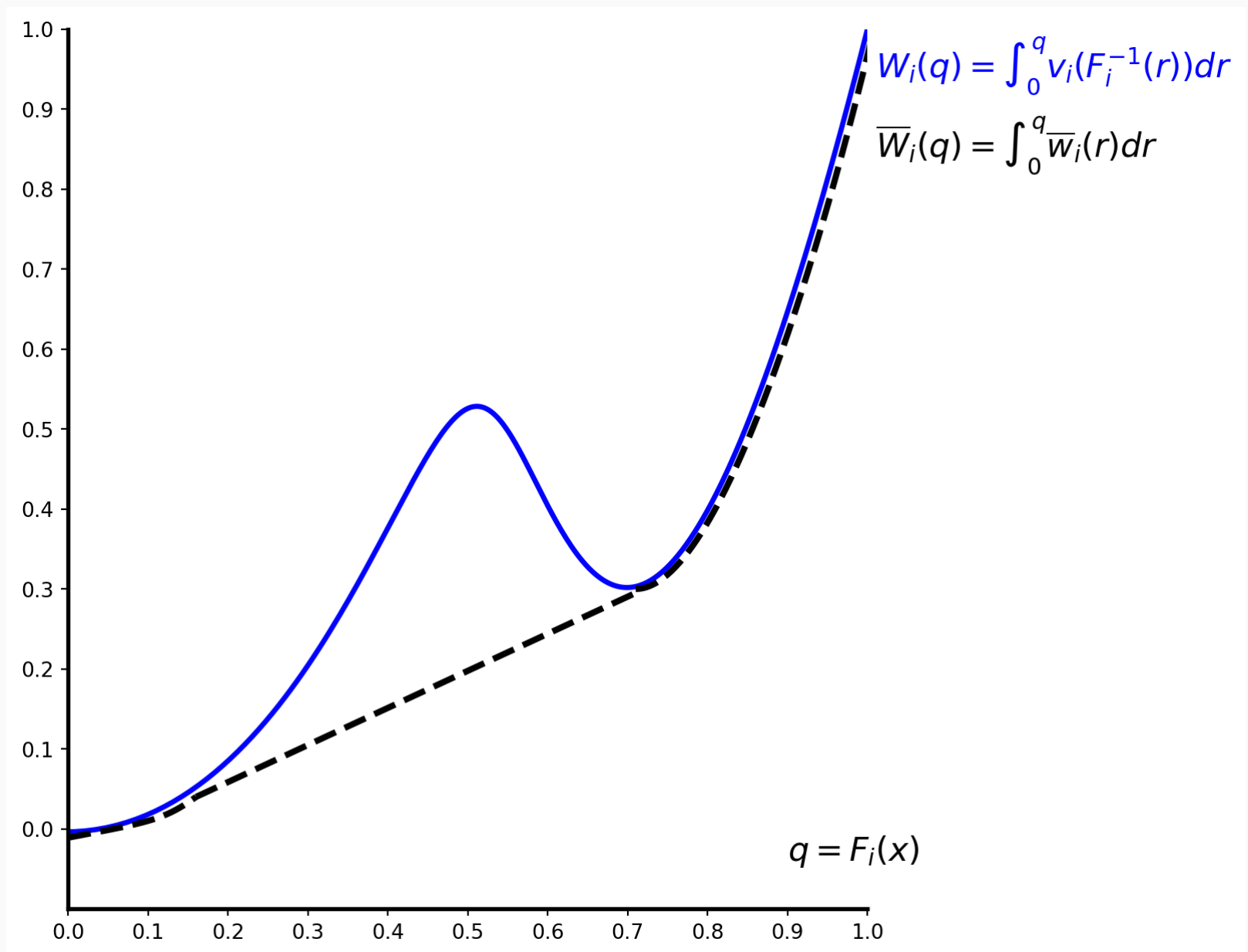
Convex hull



- Slope of green line: $\bar{v}_i([a', b']) = \frac{\int_{F_i(a')}^{F_i(b')} w(r) \, dr}{F_i(b') - F_i(a')} = \frac{\int_{a'}^{b'} v(x_i) f_i(x_i) \, dx_i}{F_i(b') - F_i(a')}$



- Slope of green line: $\bar{v}_i([a', b']) = \frac{\int_{F_i(a')}^{F_i(b')} w(r) dr}{F_i(b') - F_i(a')} = \frac{\int_{a'}^{b'} v(x_i) f_i(x_i) dx_i}{F_i(b') - F_i(a')}$



Alternative approach: linear
program, dual variable and
slack conditions

$$\mathcal{L} = \sum_{i=1}^I \langle p_i, v_i f \rangle + \langle 1, \gamma \rangle - \sum_{i=1}^I \langle p_i, \gamma \rangle + \sum_{i=1}^I \langle DE_{-i} p_i, \alpha_i \rangle$$

or, rearranging terms,

$$\mathcal{L} = \langle 1, \gamma \rangle + \sum_{i=1}^I [\langle p_i, v_i f - \gamma \rangle + \langle DE_{-i} p_i, \alpha_i \rangle],$$

where multipliers α_i, γ lie in the appropriate dual spaces.

The dual

Given the interiority assumption, a feasible $\{\bar{p}_i\}_{i=1}^I$ is a maximum of the primal if and only if $(\exists \gamma \in \prod_i^I \underline{x}_i^*, \{\alpha_i\}_{i \in I} \in [L_\infty^*([0, 1]))^I)$ such that

$$\sum_{i=1}^I \langle \bar{p}_i, v_i f \rangle = \langle 1, \gamma \rangle$$

and

$$(\forall i \in I, \forall g \in \underline{x}_i) \quad \langle g, \gamma - v_i \rangle - \langle DE_{-i}g, \alpha_i \rangle \geq 0$$

$$(\forall g \in L_\infty([0, 1]^I)_+) \quad \langle g, \gamma \rangle \geq 0$$

$$(\forall i \in I, \forall g \in L_\infty([0, 1])_+) \quad \langle g, \alpha_i \rangle \geq 0$$

Plus complementary slackness

Complementary slackness implies, in addition, that if $\{\bar{p}_i\}_{i=1}^I$ is a solution to $\{$ then

$$(\forall i \in I) \quad \langle DE_{-i}\bar{p}_i, \alpha_i \rangle = 0$$

$$(\forall i \in I) \quad \langle \bar{p}_i, v_i f - \gamma \rangle + \langle DE_{-i}\bar{p}_i, \alpha_i \rangle = 0 \text{ and}$$

$$\langle 1 - \sum_{i=1}^I \bar{p}_i, \gamma \rangle = 0.$$

$(\forall i \in I), (\forall x \in X)$ define

$$\alpha_i(x_i) = \int_{\underline{x}_i}^{x_i} [v_i(z_i) - \bar{v}_i(z_i)] f_i(z_i) \, dz_i = W_i(F_i(x_i)) - \bar{W}(F_i(x_i))$$

$$\gamma(x) = \begin{cases} [v_i(x_i) - \bar{v}_i(x_i)] f_i(x_i) f_{-i}(x_{-i}) & \text{if } \bar{p}_i(x_i, x_{-i}) > 0 \\ 0 & \text{otherwise} \end{cases}$$

$$\bar{p}_i(x) = \begin{cases} \frac{1}{|Y(x)|} & \text{if } i \in Y(x) \\ 0 & \text{otherwise} \end{cases}$$

References

- Border, Kim C. 1991. "Implementation of Reduced Form Auctions: A Geometric Approach." *Econometrica* 59 (4): 1175–87.
- Fu, Hu. 2017. "Notes on Myerson's Revenue Optimal Mechanisms." <https://fuhuthu.com/notes/notes.htm>.
- Loertscher, Simon, and Ellen V. Muir. 2025. "Optimal Labor Procurement Under Minimum Wages and Monopsony Power." <https://ellenmuir.net/papers/Loertscher-Muir-EWR-2025.pdf>.
- Mussa, Michael, and S. Rosen. 1978. "Monopoly and Product Quality." *Journal of Economic Theory* 18: 301–17.
- Rockafellar, R. Tyrrell. 1970. *Convex Analysis*. Number 28 in Princeton Mathematical Series. Princeton University Press.